

Discrete Mathematics

Complete Exam Notes — Topics 4 to 6

Abstract Algebra • Advanced Counting • Graph Theory

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1 Abstract Algebra

1.1 Axioms of Group Theory

1.1.1 What is an Algebraic Structure?

An **algebraic structure** is a set together with one or more operations that follow specific rules (axioms). We build up from the simplest structure (group) to richer ones (ring, field).

Definition: Binary Operation

A **binary operation** $*$ on a set G is a function $*$: $G \times G \rightarrow G$. It takes two elements of G and returns one element of G (this property is called **closure**).

Definition: Group

A **group** is a set G together with a binary operation $*$ satisfying four axioms:

1. **Closure:** For all $a, b \in G$, $a * b \in G$.
2. **Associativity:** For all $a, b, c \in G$, $(a * b) * c = a * (b * c)$.
3. **Identity:** There exists an element $e \in G$ such that for all $a \in G$, $e * a = a * e = a$.
4. **Inverse:** For each $a \in G$, there exists $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

If additionally $a * b = b * a$ for all $a, b \in G$, the group is called **abelian** (commutative).

Exam Alert

Memorise the four axioms with the mnemonic **CAII**: Closure, Associativity, Identity, Inverse. In an exam, always verify all four when checking if something is a group.

Example: Checking the Group Axioms

Is $(\mathbb{Z}, +)$ a group?

1. **Closure:** Sum of two integers is an integer. ✓
2. **Associativity:** $(a + b) + c = a + (b + c)$. ✓
3. **Identity:** $e = 0$, since $a + 0 = 0 + a = a$. ✓
4. **Inverse:** $a^{-1} = -a$, since $a + (-a) = 0 = e$. ✓

Also abelian since $a + b = b + a$. So $(\mathbb{Z}, +)$ is an **abelian group**. $\square$

Example: Non-Example

Is $(\mathbb{N}, +)$ a group? No — natural numbers have no inverses (there is no natural number n such that $5 + n = 0$). Axiom 4 fails.

1.2 Addition Modulo Groups

Definition: Integers Modulo n

For a positive integer n , define $\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$ with the operation **addition modulo n** : $a +_n b = (a + b) \bmod n$.

Theorem: $\mathbb{Z}_n$ under Addition is an Abelian Group

For any $n \geq 1$, $(\mathbb{Z}_n, +_n)$ satisfies all four group axioms:

- **Closure**: $(a + b) \bmod n \in \{0, \dots, n-1\}$. ✓
- **Associativity**: Inherited from integer addition. ✓
- **Identity**: $e = 0$. ✓
- **Inverse**: Inverse of a is $n - a$ (for $a \neq 0$) and 0 (for $a = 0$). ✓
- **Abelian**: $a +_n b = b +_n a$. ✓

Example: $\mathbb{Z}_5$ Addition Table

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Each row and column is a permutation of $\{0, 1, 2, 3, 4\}$ — the Latin square property.
Inverses: $0^{-1} = 0$, $1^{-1} = 4$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 1$.

1.3 Multiplication Modulo Groups

Definition: Multiplicative Group Modulo n

Define $\mathbb{Z}_n^* = \{a \in \mathbb{Z}_n \mid \gcd(a, n) = 1\}$ with operation multiplication modulo n . This is the set of all elements of $\mathbb{Z}_n$ that are *coprime* to n .

Theorem: $\mathbb{Z}_n^*$ under Multiplication is a Group

$(\mathbb{Z}_n^*, \cdot_n)$ is a group under multiplication modulo n . The identity is 1. The inverse of a is the number b such that $a \cdot b \equiv 1 \pmod{n}$ (exists iff $\gcd(a, n) = 1$, guaranteed for elements of $\mathbb{Z}_n^*$).

Example: $\mathbb{Z}_8^*$

$\mathbb{Z}_8^* = \{1, 3, 5, 7\}$ (elements coprime to 8).

Multiplication table mod 8:

$\cdot_8$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

Every element is its own inverse! This group is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Note: Euler's Totient Function

$\phi(n) = |\mathbb{Z}_n^*|$ is the **Euler totient function** — the count of integers in $\{1, \dots, n\}$ coprime to n .

- If p is prime: $\phi(p) = p - 1$.
- $\phi(p^k) = p^k - p^{k-1}$.
- $\phi(mn) = \phi(m)\phi(n)$ when $\gcd(m, n) = 1$.

1.4 Other Important Groups

Group	Set	Op	Identity	Abelian?
$(\mathbb{Z}, +)$	Integers	$+$	0	Yes
$(\mathbb{Q} \setminus \{0\}, \cdot)$	Non-zero rationals	$\cdot$	1	Yes
$(\mathbb{R} \setminus \{0\}, \cdot)$	Non-zero reals	$\cdot$	1	Yes
$(\mathbb{Z}_n, +_n)$	$\{0, \dots, n-1\}$	$+_n$	0	Yes
$(\mathbb{Z}_n^*, \cdot_n)$	Units mod n	$\cdot_n$	1	Yes
S_n (Symmetric group)	Permutations of n elements	Composition	Identity perm	No (for $n \geq 3$)
$GL_n(\mathbb{R})$	Invertible $n \times n$ matrices	Matrix mult	I_n	No (for $n \geq 2$)

1.5 Order of an Element and Order of a Group**Definition: Order of a Group**

The **order of a group** G , written $|G|$, is the number of elements in G .

Definition: Order of an Element

The **order of an element** $a \in G$, written $\text{ord}(a)$, is the *smallest positive integer* k such that $a^k = e$ (the identity).

If no such k exists, the order is infinite.

Note: Notation

a^k means $a * a * \cdots * a$ (k times). In additive notation, $a^k = ka$.

Example: Computing Orders in $\mathbb{Z}_6$

In $(\mathbb{Z}_6, +_6)$, the identity is 0. Compute $\text{ord}(2)$:

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 6 \equiv 0 \pmod{6}.$$

So $\text{ord}(2) = 3$.

Similarly: $\text{ord}(1) = 6$, $\text{ord}(2) = 3$, $\text{ord}(3) = 2$, $\text{ord}(4) = 3$, $\text{ord}(5) = 6$, $\text{ord}(0) = 1$.

Theorem: Order Divides Group Order

For any element a in a finite group G : $\text{ord}(a)$ divides $|G|$. This follows from Lagrange's Theorem (see Section 4.7).

1.6 Cyclic Groups**Definition: Generator and Cyclic Group**

An element $a \in G$ is a **generator** of G if every element of G can be written as a^k for some integer k . A group G is **cyclic** if it has at least one generator. We write $G = \langle a \rangle$.

Theorem: Basic Properties of Cyclic Groups

1. Every cyclic group is **abelian**.
2. $\mathbb{Z}_n = \langle 1 \rangle$ — the group $\mathbb{Z}_n$ is cyclic, generated by 1.
3. The element $a \in \mathbb{Z}_n$ is a generator iff $\gcd(a, n) = 1$.
4. The number of generators of $\mathbb{Z}_n$ is $\phi(n)$.
5. Every subgroup of a cyclic group is itself cyclic.

Example: Finding Generators of $\mathbb{Z}_{10}$

$\phi(10) = \phi(2)\phi(5) = 1 \cdot 4 = 4$, so there are 4 generators.
Generators are elements a with $\gcd(a, 10) = 1$: $\{1, 3, 7, 9\}$.

Theorem: Cyclic Groups are Classified

- Every infinite cyclic group is isomorphic to $(\mathbb{Z}, +)$.
- Every finite cyclic group of order n is isomorphic to $(\mathbb{Z}_n, +_n)$.

1.7 Lagrange's Theorem and Corollaries

Definition: Subgroup

A subset $H \subseteq G$ is a **subgroup** of G (written $H \leq G$) if H is itself a group under the same operation. To verify, use the **subgroup test**:

- H is non-empty.
- For all $a, b \in H$: $a * b^{-1} \in H$ (closed under operation and inverses).

Definition: Coset

Let $H \leq G$ and $a \in G$. The **left coset** of H with representative a is:

$$aH = \{a * h \mid h \in H\}$$

Similarly, the right coset is $Ha = \{h * a \mid h \in H\}$.

Theorem: Lagrange's Theorem

Let G be a finite group and $H \leq G$ a subgroup. Then:

$$|H| \text{ divides } |G|$$

More precisely, $|G| = |H| \cdot [G : H]$ where $[G : H]$ is the number of distinct left cosets of H in G (called the **index** of H in G).

Example: Lagrange's Theorem in Practice

$G = \mathbb{Z}_6$, $|G| = 6$. Possible subgroup sizes are the divisors of 6: $\{1, 2, 3, 6\}$.
Indeed: $H_1 = \{0\}$ (size 1), $H_2 = \{0, 3\}$ (size 2), $H_3 = \{0, 2, 4\}$ (size 3), $H_4 = \mathbb{Z}_6$ (size 6).
There is **no subgroup of size 4 or 5**, since 4 and 5 do not divide 6.

Theorem: Corollaries of Lagrange's Theorem

1. **Order of element divides order of group:** $\text{ord}(a)$ divides $|G|$ for all $a \in G$.
2. **Groups of prime order are cyclic:** If $|G| = p$ (prime), then G is cyclic and every non-identity element is a generator.
3. **Fermat's Little Theorem:** If p is prime and $\gcd(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$.
4. **Euler's Theorem:** If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Example: Using Fermat's Little Theorem

Compute $2^{100} \bmod 7$.

Since 7 is prime: $2^6 \equiv 1 \pmod{7}$.

$100 = 6 \cdot 16 + 4$, so $2^{100} = (2^6)^{16} \cdot 2^4 \equiv 1^{16} \cdot 16 \equiv 16 \bmod 7 \equiv 2 \pmod{7}$.

1.8 Axioms of Ring Theory**Definition: Ring**

A **ring** is a set R with two binary operations $+$ (addition) and $\cdot$ (multiplication) satisfying:

1. $(R, +)$ is an **abelian group** (with identity 0).
2. **Associativity of multiplication:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
3. **Distributivity:** $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

If also $a \cdot b = b \cdot a$ for all a, b , it is a **commutative ring**.

If there exists a multiplicative identity 1 with $1 \cdot a = a \cdot 1 = a$, it is a **ring with unity**.

Example: Examples of Rings

- $(\mathbb{Z}, +, \cdot)$ — integers. Commutative ring with unity.
- $(\mathbb{Z}_n, +_n, \cdot_n)$ — integers mod n . Commutative ring with unity.
- $M_{n \times n}(\mathbb{R})$ — $n \times n$ real matrices. Ring with unity, but *not* commutative.
- $\{0\}$ — trivial ring (zero ring).

Definition: Integral Domain and Zero Divisors

A **zero divisor** in a ring is a non-zero element a such that $a \cdot b = 0$ for some non-zero b .

A commutative ring with unity and **no zero divisors** is called an **integral domain**.

Example: Zero Divisors in $\mathbb{Z}_6$

$2 \cdot 3 = 6 \equiv 0 \pmod{6}$. So 2 and 3 are zero divisors in $\mathbb{Z}_6$.

$\mathbb{Z}_6$ is **not** an integral domain. However, $\mathbb{Z}_p$ for prime p **is** an integral domain (and even a field).

1.9 Polynomial Rings

Definition: Polynomial Ring $R[x]$

Let R be a ring. The **polynomial ring** $R[x]$ is the set of all polynomials with coefficients in R :

$$R[x] = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \mid a_i \in R, n \geq 0\}$$

with the usual polynomial addition and multiplication.

Theorem: Properties of Polynomial Rings

- If R is a commutative ring, so is $R[x]$.
- If R is an integral domain, so is $R[x]$.
- Degree: $\deg(f \cdot g) = \deg(f) + \deg(g)$ (over an integral domain).
- **Division Algorithm:** For $f, g \in F[x]$ with F a field and $g \neq 0$, there exist unique $q, r \in F[x]$ with $f = gq + r$ and $\deg(r) < \deg(g)$.

Definition: Irreducible Polynomial

A polynomial $p(x) \in F[x]$ of degree ≥ 1 is **irreducible** over F if it cannot be factored as a product of two polynomials of lower degree with coefficients in F . (Analogous to prime numbers.)

Example: Irreducible over $\mathbb{Z}_2$

$x^2 + x + 1$ over $\mathbb{Z}_2$: test $x = 0$: $0 + 0 + 1 = 1 \neq 0$; $x = 1$: $1 + 1 + 1 = 3 \equiv 1 \neq 0$. No roots in $\mathbb{Z}_2$, so it is **irreducible** over $\mathbb{Z}_2$.

1.10 Axioms of Fields

Definition: Field

A **field** is a commutative ring with unity $(F, +, \cdot)$ in which every non-zero element has a multiplicative inverse:

$$\forall a \in F, a \neq 0 \Rightarrow \exists a^{-1} \in F : a \cdot a^{-1} = 1$$

Equivalently: $(F, +)$ is an abelian group, $(F \setminus \{0\}, \cdot)$ is an abelian group, and multiplication distributes over addition.

Structure	Extra axiom over ring	Examples
Ring	—	$\mathbb{Z}, \mathbb{Z}_n, M_n(\mathbb{R})$
Commutative Ring	$ab = ba$	$\mathbb{Z}, \mathbb{Z}_n, \mathbb{Q}[x]$
Integral Domain	No zero divisors	$\mathbb{Z}, \mathbb{Z}_p$ (p prime)
Field	Every non-zero has mult. inverse	$\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p$

Theorem: Finite Fields (Galois Fields)

A finite field of order q exists if and only if $q = p^n$ for some prime p and $n \geq 1$. It is written $\mathbb{F}_q$ or $GF(q)$.

Examples: $GF(2) = \mathbb{Z}_2$, $GF(4) = \mathbb{Z}_2[x]/(x^2 + x + 1)$, $GF(8) = \mathbb{Z}_2[x]/(x^3 + x + 1)$.

1.11 Primitive Elements**Definition: Primitive Element (Primitive Root)**

Let $G = (F^*, \cdot)$ be the multiplicative group of a finite field $F^* = F \setminus \{0\}$. A **primitive element** (or **primitive root**) $g \in F^*$ is a generator of this cyclic group:

$$F^* = \langle g \rangle = \{g^0, g^1, g^2, \dots, g^{q-2}\}$$

Theorem: Multiplicative Group of a Finite Field is Cyclic

For any finite field $GF(q)$, the multiplicative group $(GF(q)^*, \cdot)$ is cyclic. Therefore, primitive elements always exist.

Example: Primitive Root of $\mathbb{Z}_7^*$

$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$, $|\mathbb{Z}_7^*| = 6$.

Check $g = 3$: $3^1 = 3$, $3^2 = 2$, $3^3 = 6$, $3^4 = 4$, $3^5 = 5$, $3^6 = 1 \pmod{7}$.

All 6 elements of $\mathbb{Z}_7^*$ appear, so 3 is a primitive root (generator) of $\mathbb{Z}_7^*$.

1.12 Applications in Cryptography**1.12.1 The Discrete Logarithm Problem****Definition: Discrete Logarithm Problem (DLP)**

Given a cyclic group G with generator g , a prime p , and an element $y \in G$, find the integer x such that:

$$g^x \equiv y \pmod{p}$$

x is called the **discrete logarithm** of y to base g modulo p .

Note: Why is DLP Hard?

Forward direction (easy): Given g, x, p , compute $g^x \pmod{p}$ quickly using *fast exponentiation* (square-and-multiply) in $O(\log x)$ multiplications.

Backward direction (hard): Given g, y, p , find x . No efficient classical algorithm is known for large p . This computational hardness is the foundation of many cryptographic protocols.

1.12.2 Diffie-Hellman Key Exchange

Definition: Diffie-Hellman Key Exchange Protocol

Allows two parties (Alice and Bob) to establish a **shared secret** over a public channel without ever transmitting the secret itself.

Setup (public): A large prime p and a primitive root g modulo p are published.

Protocol steps:

1. **Alice** picks a secret integer a , computes $A = g^a \bmod p$, and sends A to Bob.
2. **Bob** picks a secret integer b , computes $B = g^b \bmod p$, and sends B to Alice.
3. **Alice** computes the shared key: $K = B^a \bmod p = g^{ab} \bmod p$.
4. **Bob** computes the shared key: $K = A^b \bmod p = g^{ab} \bmod p$.

Both arrive at the same $K = g^{ab} \bmod p$.

Example: Diffie-Hellman (Small Numbers)

Public: $p = 23$, $g = 5$.

Alice: $a = 6$, sends $A = 5^6 \bmod 23 = 15625 \bmod 23 = 8$.

Bob: $b = 15$, sends $B = 5^{15} \bmod 23 = 19$.

Alice: $K = 19^6 \bmod 23 = 2$.

Bob: $K = 8^{15} \bmod 23 = 2$. ✓ Shared secret is $K = 2$.

Key Point

An eavesdropper sees p, g, A, B but must solve the DLP to find a or b . For p having 2048 bits, this is computationally infeasible with current technology.

2 Advanced Counting

2.1 Recurrence Relations of Recursion Algorithms

2.1.1 What is a Recurrence Relation?

Definition: Recurrence Relation

A **recurrence relation** defines each term of a sequence in terms of earlier terms. For algorithm analysis, $T(n)$ is the running time on input of size n , expressed as a function of T on smaller inputs.

Example: Factorial

factorial(n): if $n = 0$ return 1, else return $n \cdot \text{factorial}(n - 1)$.

Recurrence: $T(n) = T(n - 1) + c$ (one recursive call + $O(1)$ work), with $T(0) = c_0$.

Solution: $T(n) = nc + c_0 = O(n)$.

Example: Tower of Hanoi

Moving n disks: move top $n - 1$ to spare peg ($T(n - 1)$ moves), move bottom disk (1 move), move $n - 1$ back ($T(n - 1)$ moves).

Recurrence: $T(n) = 2T(n - 1) + 1$, $T(1) = 1$.

Solution by iteration:

$$\begin{aligned} T(n) &= 2T(n - 1) + 1 \\ &= 2(2T(n - 2) + 1) + 1 = 4T(n - 2) + 3 \\ &= 8T(n - 3) + 7 = \dots = 2^{n-1}T(1) + 2^{n-1} - 1 = 2^n - 1. \end{aligned}$$

2.1.2 Solving Linear Recurrences

Definition: Linear Homogeneous Recurrence with Constant Coefficients

Form: $T(n) = c_1T(n - 1) + c_2T(n - 2) + \dots + c_kT(n - k)$.

To solve, find the **characteristic equation**:

$$r^k = c_1r^{k-1} + c_2r^{k-2} + \dots + c_k$$

Theorem: Solution from Characteristic Roots

- If the characteristic equation has k **distinct** roots $r_1, r_2, \dots, r_k$:

$$T(n) = A_1r_1^n + A_2r_2^n + \dots + A_kr_k^n$$

- If root r_i has **multiplicity** m : it contributes terms $(A_0 + A_1n + \dots + A_{m-1}n^{m-1})r_i^n$.
- Constants A_i are determined by **initial conditions**.

Example: Fibonacci Numbers

$$F(n) = F(n-1) + F(n-2), F(0) = 0, F(1) = 1.$$

Characteristic equation: $r^2 = r + 1 \Rightarrow r^2 - r - 1 = 0$.

Roots: $r = \frac{1 \pm \sqrt{5}}{2}$. Let $\phi = \frac{1 + \sqrt{5}}{2}$ (golden ratio), $\psi = \frac{1 - \sqrt{5}}{2}$.

General solution: $F(n) = A\phi^n + B\psi^n$.

From $F(0) = 0$: $A + B = 0 \Rightarrow B = -A$.

From $F(1) = 1$: $A\phi - A\psi = 1 \Rightarrow A(\phi - \psi) = A\sqrt{5} = 1 \Rightarrow A = \frac{1}{\sqrt{5}}$.

Binet's Formula: $F(n) = \frac{\phi^n - \psi^n}{\sqrt{5}}$.

2.2 Recurrence Relations of Divide and Conquer Algorithms**Definition: Divide and Conquer Recurrence**

When an algorithm divides a problem of size n into a subproblems of size n/b , and does $f(n)$ work outside the recursive calls:

$$T(n) = aT(n/b) + f(n), \quad T(1) = c$$

Example: Merge Sort

Divides array in half ($a = 2$, $b = 2$), merges in $O(n)$:

$$T(n) = 2T(n/2) + O(n)$$

Solution: $T(n) = O(n \log n)$.

Example: Binary Search

Checks midpoint ($O(1)$), recurses on one half ($a = 1$, $b = 2$):

$$T(n) = T(n/2) + O(1)$$

Solution: $T(n) = O(\log n)$.

2.3 Master's Theorem and Growth of Functions**Theorem: Master Theorem**

For $T(n) = aT(n/b) + f(n)$ with $a \geq 1$, $b > 1$:

1. **Case 1:** If $f(n) = O(n^{\log_b a - \varepsilon})$ for some $\varepsilon > 0$:

$$T(n) = \Theta(n^{\log_b a})$$

(Recursive work dominates.)

2. **Case 2:** If $f(n) = \Theta(n^{\log_b a})$:

$$T(n) = \Theta(n^{\log_b a} \log n)$$

(Both parts are equal.)

3. **Case 3:** If $f(n) = \Omega(n^{\log_b a + \varepsilon})$ for some $\varepsilon > 0$, and $af(n/b) \leq cf(n)$ for some $c < 1$:

$$T(n) = \Theta(f(n))$$

(Non-recursive work dominates.)

Example: Applying the Master Theorem

Recurrence	a	b	$\log_b a$	Case & Result
$T(n) = 2T(n/2) + n$	2	2	1	Case 2: $\Theta(n \log n)$
$T(n) = 2T(n/2) + 1$	2	2	1	Case 1: $\Theta(n)$
$T(n) = 4T(n/2) + n$	4	2	2	Case 1: $\Theta(n^2)$
$T(n) = T(n/2) + n$	1	2	0	Case 3: $\Theta(n)$
$T(n) = 3T(n/3) + n \log n$	3	3	1	Case 2*: $\Theta(n \log^2 n)$

*Extended Case 2 with log factors.

Exam Alert

The Master Theorem does NOT apply when: (a) $a < 1$ or $b \leq 1$, (b) $f(n)$ is not polynomially larger/smaller than $n^{\log_b a}$ (e.g. $f(n) = n^{\log_b a} / \log n$ falls between cases 1 and 2). In such cases, use substitution or the Akra-Bazzi method.

2.4 Generating Functions

Definition: Ordinary Generating Function

The **ordinary generating function (OGF)** of a sequence $a_0, a_1, a_2, \dots$ is the formal power series:

$$G(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$

We treat x as a formal variable; convergence is not required.

Note: Key OGF Pairs to Memorise

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$$

$$\frac{1}{1-ax} = \sum_{n=0}^{\infty} a^n x^n$$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n$$

$$\frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} \binom{n+k-1}{k-1} x^n$$

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

2.5 Applications of Generating Functions in Counting**Theorem: Coefficient Extraction**

The number of ways to count objects with given constraints is often the coefficient $[x^n] G(x)$ of a generating function.

Example: Distributing Identical Objects

How many ways to distribute n identical balls into 3 distinct boxes?

Each box can receive $0, 1, 2, \dots$ balls. The GF for one box is $\frac{1}{1-x}$.

For 3 boxes: $G(x) = \left(\frac{1}{1-x}\right)^3 = \frac{1}{(1-x)^3} = \sum_{n=0}^{\infty} \binom{n+2}{2} x^n$.

Answer: $[x^n] G(x) = \binom{n+2}{2}$.

Example: Restricted Distributions

How many ways to distribute n identical balls into 3 boxes with each box receiving *at least 1* ball?

Each box: $x + x^2 + x^3 + \dots = \frac{x}{1-x}$.

$G(x) = \left(\frac{x}{1-x}\right)^3 = \frac{x^3}{(1-x)^3}$.

$[x^n] G(x) = [x^{n-3}] \frac{1}{(1-x)^3} = \binom{n-3+2}{2} = \binom{n-1}{2}$.

2.6 Applications of Generating Functions in Solving Recurrence Relations

Example: Solving a Recurrence with GFs

Solve $a_n = 3a_{n-1} + 2$, $a_0 = 1$.

Step 1: Multiply both sides by x^n and sum over $n \geq 1$:

$$\sum_{n \geq 1} a_n x^n = 3x \sum_{n \geq 1} a_{n-1} x^{n-1} + 2 \sum_{n \geq 1} x^n$$

$$G(x) - a_0 = 3x G(x) + \frac{2x}{1-x}$$

Step 2: Solve for $G(x)$:

$$G(x)(1-3x) = 1 + \frac{2x}{1-x} = \frac{1-x+2x}{1-x} = \frac{1+x}{1-x}$$

$$G(x) = \frac{1+x}{(1-x)(1-3x)}$$

Step 3: Partial fractions: $\frac{1+x}{(1-x)(1-3x)} = \frac{A}{1-x} + \frac{B}{1-3x}$.

Solving: $A = -1$, $B = 2$. So:

$$G(x) = \frac{-1}{1-x} + \frac{2}{1-3x} = -\sum x^n + 2 \sum 3^n x^n$$

$$a_n = 2 \cdot 3^n - 1$$

Check: $a_0 = 2 - 1 = 1$. ✓ $a_1 = 3(1) + 2 = 5 = 2(3) - 1 = 5$. ✓

Key Point

General strategy for GF solution of recurrences:

1. Define $G(x) = \sum a_n x^n$.
2. Multiply the recurrence by x^n and sum over relevant n .
3. Use initial conditions to get a closed form for $G(x)$.
4. Expand $G(x)$ via partial fractions or known series to read off a_n .

3 Graph Theory

3.1 Graph Terminologies

Definition: Graph

A **graph** $G = (V, E)$ consists of:

- A non-empty set V of **vertices** (also called nodes).
- A set E of **edges**, where each edge connects two vertices.

The number of vertices $|V|$ is the **order** of the graph; $|E|$ is its **size**.

Term	Meaning
Adjacent / Neighbours	Two vertices connected by an edge
Incident	A vertex and an edge are incident if the vertex is an endpoint
Degree $\deg(v)$	Number of edges incident to v
Loop	An edge from a vertex to itself
Multiple edges	More than one edge between the same pair of vertices
Simple graph	No loops, no multiple edges
Walk	A sequence of vertices and edges (may repeat)
Trail	A walk with no repeated edge
Path	A walk with no repeated vertex
Cycle	A closed path (starts and ends at same vertex, no repeats)
Connected	A path exists between every pair of vertices
Component	Maximal connected subgraph

Theorem: Handshaking Lemma

For any graph $G = (V, E)$:

$$\sum_{v \in V} \deg(v) = 2|E|$$

Proof: Each edge contributes exactly 2 to the total degree sum (one for each endpoint).

Corollary 3.1. *The number of vertices with **odd degree** is always even.*

Example: Handshaking

A graph has 6 vertices with degrees 1, 2, 3, 3, 4, 5. Sum = 18 = 2×9 , so it has 9 edges. Odd-degree vertices: 1, 3, 3, 5 — four of them (even count). ✓

3.2 Types of Graphs

Type	Description
Simple Graph	No loops, no multiple edges
Multigraph	Allows multiple edges (but no loops)
Pseudograph	Allows both loops and multiple edges
Directed Graph (Digraph)	Edges have direction (ordered pairs)
Weighted Graph	Each edge has an associated numerical weight
Undirected Graph	Edges have no direction
Complete Graph K_n	Every pair of vertices is connected
Bipartite Graph	Vertices split into two sets; edges only go between sets
Tree	Connected, acyclic graph
Forest	Acyclic graph (possibly disconnected)
Planar Graph	Can be drawn in a plane with no edge crossings

3.2.1 Subgraphs

Definition: Subgraph

$H = (V', E')$ is a **subgraph** of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$ with all endpoints of E' in V' .

A **spanning subgraph** has $V' = V$ (uses all vertices).

An **induced subgraph** $G[V']$ contains all edges of G between vertices in V' .

3.3 Some Common Graphs

3.3.1 Complete Graph K_n

Definition: K_n — Complete Graph

K_n has n vertices and an edge between **every** pair of vertices.

$$|V| = n, \quad |E| = \binom{n}{2} = \frac{n(n-1)}{2}$$

Every vertex has degree $n - 1$. (K_n is $(n - 1)$ -regular.)

Graph	Vertices	Edges
K_1	1	0
K_2	2	1
K_3	3	3
K_4	4	6
K_5	5	10
K_n	n	$n(n-1)/2$

3.3.2 Cycle Graph C_n

Definition: C_n — Cycle Graph

C_n has n vertices arranged in a single cycle.

$$|V| = n, \quad |E| = n \quad (n \geq 3)$$

Every vertex has degree 2. (C_n is 2-regular.)

3.3.3 Wheel Graph W_n

Definition: W_n — Wheel Graph

W_n is obtained by adding a single **hub vertex** connected to all vertices of C_n .

$$|V| = n + 1, \quad |E| = 2n$$

The hub has degree n ; the rim vertices have degree 3.

3.3.4 Hypercube Graph Q_n

Definition: Q_n — n -dimensional Hypercube

Q_n has one vertex for each n -bit binary string. Two vertices are adjacent iff their binary labels differ in **exactly one bit**.

$$|V| = 2^n, \quad |E| = n \cdot 2^{n-1}$$

Every vertex has degree n (Q_n is n -regular).

Example: Q_3

Vertices: 000, 001, 010, 011, 100, 101, 110, 111 (8 vertices).

000 is adjacent to 001, 010, 100 (differ in one bit). Each vertex has degree 3.

$$|E| = 3 \times 2^2 = 12.$$

3.3.5 Complete Bipartite Graph $K_{m,n}$

Definition: $K_{m,n}$ — Complete Bipartite Graph

$K_{m,n}$ has vertex sets V_1 (size m) and V_2 (size n) with an edge between *every* vertex in V_1 and *every* vertex in V_2 (no edges within the same set).

$$|V| = m + n, \quad |E| = mn$$

Vertices in V_1 have degree n ; vertices in V_2 have degree m .

Graph	Vertices	Edges
$K_{1,n}$ (Star)	$n + 1$	n
$K_{2,3}$	5	6
$K_{3,3}$	6	9
$K_{m,n}$	$m + n$	mn

3.4 Bipartite Graphs

Definition: Bipartite Graph

A graph $G = (V, E)$ is **bipartite** if the vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge has one endpoint in V_1 and the other in V_2 .

Theorem: Characterisation of Bipartite Graphs

A graph is bipartite if and only if it contains **no odd-length cycle**.

Intuition: In a bipartite graph you alternate between V_1 and V_2 as you traverse a cycle. To return to the start, you need an even number of steps.

Example: Testing Bipartiteness (2-Colouring)

Algorithm: BFS/DFS with 2 colours. Assign one colour to the start vertex; assign the opposite colour to each neighbour. If a conflict arises (adjacent vertices get the same colour), the graph is **not** bipartite; otherwise it **is** bipartite.

Example: Is C_6 Bipartite?

C_6 (even cycle): $V_1 = \{v_1, v_3, v_5\}$, $V_2 = \{v_2, v_4, v_6\}$. Every edge goes $V_1 \leftrightarrow V_2$. Yes, bipartite. ✓

C_5 (odd cycle): Cannot 2-colour without conflict. **Not** bipartite.

Note: Common Bipartite Graphs

$K_{m,n}$, all trees, C_{2k} (even cycles), Q_n (hypercubes).

All of K_3, C_5, K_4 (non-bipartite): contain odd cycles.

3.5 Euler Paths and Hamiltonian Paths

3.5.1 Euler Paths and Circuits

Definition: Euler Path and Circuit

- An **Euler path** (or Eulerian trail) is a **trail** (no repeated edges) that visits **every edge** exactly once.
- An **Euler circuit** is an Euler path that starts and ends at the **same vertex**.

Theorem: Euler Circuit Theorem — Euler 1736

A connected graph G has an **Euler circuit** if and only if **every vertex has even degree**.

Theorem: Euler Path Theorem

A connected graph G has an **Euler path** (but not a circuit) if and only if it has **exactly two vertices of odd degree**. The path starts at one odd-degree vertex and ends at the other.

Example: The Königsberg Bridge Problem

The original problem (Euler, 1736): can you cross all 7 bridges of Königsberg exactly once?

Model as a graph: 4 landmasses (vertices) and 7 bridges (edges).

Degrees: 3, 3, 3, 5 — all four vertices have **odd degree**.

For an Euler path we need exactly 2 odd-degree vertices; for an Euler circuit, 0.

Having 4 odd-degree vertices means **no Euler path exists**. It is impossible.

Example: Deciding Euler Path/Circuit

Graph	Odd-degree vertices	Euler Circuit?	Euler Path?
K_4	0 (all degree 3... wait, degree is 3 odd!)	No	No (4 odd)
K_4 corrected	all have degree 3: 4 odd-degree vertices	No	No
K_5	0 (all degree 4)	Yes	—
C_n	0 (all degree 2)	Yes	—
$K_{2,3}$	4 odd-degree vertices	No	No
Path P_n	2 (the two endpoints, degree 1)	No	Yes

3.5.2 Fleury's Algorithm (Finding an Euler Circuit)

1. Start at any vertex (any odd-degree vertex if finding an Euler path).
2. At each step, choose an edge that is **not a bridge** (an edge whose removal disconnects the graph), unless there is no other choice.
3. Remove the traversed edge. Continue until all edges are used.

3.5.3 Hamiltonian Paths and Circuits**Definition: Hamiltonian Path and Circuit**

- A **Hamiltonian path** is a path that visits **every vertex** exactly once.
- A **Hamiltonian circuit** is a Hamiltonian path that starts and ends at the **same vertex**.

Exam Alert

Unlike Euler paths (which have a clean necessary and sufficient condition), there is **no known efficient characterisation** of Hamiltonian graphs. Deciding whether a Hamiltonian circuit exists is an **NP-complete** problem.

Theorem: Dirac's Theorem (Sufficient Condition)

If G is a simple graph with $n \geq 3$ vertices and every vertex has degree $\deg(v) \geq n/2$, then G has a Hamiltonian circuit.

Theorem: Ore's Theorem (Sufficient Condition)

If G is a simple graph with $n \geq 3$ vertices and for every pair of *non-adjacent* vertices u, v : $\deg(u) + \deg(v) \geq n$, then G has a Hamiltonian circuit.

Note: These are Sufficient Conditions Only — Not Necessary

A graph can have a Hamiltonian circuit even if neither Dirac's nor Ore's condition holds. These theorems only guarantee existence when the condition is met; they cannot be used to prove non-existence.

3.5.4 Euler vs Hamiltonian: Key Comparison

Property	Euler	Hamiltonian
Visits	Every edge once	Every vertex once
Key condition	Vertex degrees (even/odd)	No clean condition
Characterisation	Clean (Euler's theorem)	None (NP-complete to decide)
Algorithm	Efficient (Fleury's, Hierholzer's)	No efficient algorithm known
Circuit condition	All degrees even	No known simple condition
Path condition	Exactly 2 odd-degree vertices	No known simple condition

Example: Hamiltonian Circuit in K_n

K_n always has a Hamiltonian circuit for $n \geq 3$: just visit all vertices in any order and return to start. The number of distinct Hamiltonian circuits in K_n is $\frac{(n-1)!}{2}$.

Example: No Hamiltonian Circuit in $K_{1,3}$ (Star)

$K_{1,3}$ has a centre vertex connected to 3 leaves. Any path starting from a leaf must go through the centre, then to another leaf — but then it cannot visit the third leaf and return to start without revisiting the centre. So $K_{1,3}$ has **no Hamiltonian circuit**.

3 Quick Revision Summary (Topics 4–6)

Topic 4: Abstract Algebra

- Group axioms: **CAII** — Closure, Associativity, Identity, Inverse.
- Abelian group adds commutativity. $(\mathbb{Z}_n, +_n)$ is always an abelian group.
- $\mathbb{Z}_n^*$ = elements coprime to n ; $|\mathbb{Z}_n^*| = \phi(n)$ (Euler totient).
- Order of element a : smallest k with $a^k = e$. Divides $|G|$ (Lagrange).
- Cyclic group: generated by one element. $\mathbb{Z}_n$ is cyclic; generators are elements coprime to n .
- **Lagrange**: $|H|$ divides $|G|$ for any subgroup $H \leq G$.
- **Fermat's Little Theorem**: $a^{p-1} \equiv 1 \pmod{p}$ (p prime, $\gcd(a, p) = 1$).
- Ring: abelian group under $+$, associative $\cdot$, distributive. Field: ring where every non-zero element has a mult. inverse.
- Irreducible polynomial: like a prime in polynomial ring.
- Primitive root g : generates all of $\mathbb{Z}_p^*$. DLP: find x given $g^x \equiv y \pmod{p}$.
- Diffie-Hellman: Alice sends g^a , Bob sends g^b ; shared key = g^{ab} .

Topic 5: Advanced Counting

- Recurrence from recursion: one call of size $n-1$ gives $T(n) = T(n-1) + f(n)$.
- Characteristic equation for linear homogeneous recurrences: substitute r^n .
- Fibonacci: $F(n) = F(n-1) + F(n-2)$; closed form via Binet's formula.
- D&C recurrence: $T(n) = aT(n/b) + f(n)$. Solve with Master Theorem.
- Master Theorem: compare $f(n)$ with $n^{\log_b a}$. Three cases: Case 1 (recursive dominates), Case 2 (equal), Case 3 (non-recursive dominates).
- OGF: $G(x) = \sum a_n x^n$. Key series: $\frac{1}{1-x}$, $\frac{1}{(1-x)^k}$, $(1+x)^n$.
- GF for counting: multiply GFs of independent choices; read off coefficient $[x^n]$.
- GF for recurrences: write $G(x)$, use recurrence to form equation, solve, expand via partial fractions.

Topic 6: Graph Theory

- Graph $G = (V, E)$. Handshaking: $\sum \deg(v) = 2|E|$; odd-degree count is even.
- K_n : $n(n-1)/2$ edges, $(n-1)$ -regular. C_n : n edges, 2-regular. W_n : $n+1$ vertices, $2n$ edges. Q_n : 2^n vertices, $n \cdot 2^{n-1}$ edges, n -regular. $K_{m,n}$: mn edges.
- Bipartite $\Leftrightarrow$ no odd cycle $\Leftrightarrow$ 2-colourable. Test with BFS.
- **Euler circuit**: exists iff connected + all degrees even.
- **Euler path**: exists iff connected + exactly 2 odd-degree vertices.
- Königsberg: 4 odd-degree vertices $\Rightarrow$ neither Euler path nor circuit.
- **Hamiltonian**: visits every vertex once. No clean characterisation (NP-complete).
- Dirac: $\deg(v) \geq n/2$ for all $v \Rightarrow$ Hamiltonian circuit exists.
- Ore: $\deg(u) + \deg(v) \geq n$ for all non-adjacent $u, v \Rightarrow$ Hamiltonian circuit exists.